

# Embedding derived animacy rankings offer insights into the sources of grammatical animacy

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## Abstract

In this study, we applied the semantic projection approach to animacy, a feature that has not been previously explored using this method. We compared the relative animacy rankings of nouns denoting animals, humans, objects, and first-, second-, and third-person pronouns, as derived from word embeddings, with rankings derived from human behavioral ratings of animacy and from grammatical patterns. Our results support the semantic projection approach as an effective method for deriving proxies of human perception from word embeddings and offer insights into the sources of grammatical animacy.

## 1 Introduction

Why are grammars the way they are? One perspective—advocated to varying degrees by proponents of usage-based linguistics, cognitive linguistics, emergentism, and other functionally oriented approaches (Bybee, 2023, a.o.)—argues that grammatical structures emerge from exposure and interaction. According to this view, language learners develop grammars that align with existing linguistic norms because those are the structures they encounter. When learners’ grammars diverge from established norms, these deviations are often attributed to limited or skewed linguistic input (e.g., in L1 and L2 learning, or contact-induced change). Consequently, even when linguistic patterns conflict with cognitive or perceptual expectations, they are still learned. A compelling example is grammatical gender: any fluent speaker of German would classify the noun *Mädchen* (‘girl’) as grammatically neuter, despite girls being biologically and conceptually feminine. Studies on frequency effects and exemplar-based learning further support the role of distributional patterns in shaping both diachronic language change and real-time language processing.

A contrasting perspective, most prominently associated with generative linguistics (Chomsky, 1965, a.o.), posits that grammar is shaped by innate cognitive biases. According to this view, the acquisition of grammar is guided by inductive biases—cognitive predispositions that shape how learners interpret and generalize from linguistic data.

In a naturalistic setting of language acquisition (as opposed to an unnatural/artificial setting where learners are exposed to inconsistent input, as in artificial language learning experiments), where learners’ exposure generally aligns with the target grammar (e.g., in the case of grammatical gender in German), a functionalist explanation alone may seem sufficient. Finding support for inductive biases would crucially involve identifying cases where linguistic input and grammatical structures diverge. Such cases provide strong evidence that learners internalize grammars that do not simply mirror the distributional properties of their linguistic environment. In this study, we argue that grammatical animacy represents such a case. Our contributions are summarized as follows:

- Methodologically, we conduct extensive experiments to capture the representation of animacy in English. The application of the semantic projection method to animacy is novel, as animacy has not been systematically examined using this approach.
- Empirically, we demonstrate that semantic projections derived from English word embeddings align closely with human perceptions of animacy, as reflected in animacy rankings based on human ratings.
- We further show that the distributional patterns of linguistic data in English—captured through embedding-derived animacy rankings—do not fully align with grammatical

| Animate     |            |           |
|-------------|------------|-----------|
| Human       | Nonhuman   | Inanimate |
| <i>-man</i> | <i>-ma</i> | -         |

Table 1: Animacy modulates plural marking in the Gwandji dialect of Wambaya (Aguas, 1968:5-6; cited in Santazilia, 2020: Table 7)

patterns as reflected in the grammar-based animacy hierarchy. This finding suggests that animacy as a grammatical feature is likely influenced by innate cognitive biases, enabling first-language learners to override distributional tendencies in their input.

## 2 Related works

### 2.1 The semantic projection method

Word embeddings have been shown to capture semantic relations through analogy (Mikolov et al., 2013). However, these captured relations are typically binary, meaning that they reflect definitive or widely agreed-upon mappings between word pairs. For example, the pair *king* and *queen* represents a "male-female" gender contrast, and by analogy, the female counterpart for *waiter* would be *waitress*, which is generally considered the optimal, if not the only, correct answer. Similarly, in the relation country-capital, as in *Germany-Berlin*, a prompt like *China* would be expected to elicit *Beijing* and no other city.

Yet, many semantic relations are scalar in nature and do not necessarily involve such definitive oppositions between word pairs. For instance, along the semantic dimension of "size," many things are larger than bees. In the analogy *ant: bee, bee: ?*, we might be asking for an entity that is as many times larger than a bee as a bee is larger than an ant. The answer, however, is not unique: cicadas, mantises, or grasshoppers might all be appropriate responses.

Grand et al. (2022) introduced the semantic projection method, which extracts vectors from word embeddings to represent semantic relations or features such as size or danger level. The authors demonstrated that words' projections onto these vectors correlated with human ratings on 17 semantic features. Thus, the semantic projection method appears to be a reliable proxy for human perception of various semantic dimensions. However, animacy was not included among the features they investigated, presenting an important gap.

### 2.2 Animacy hierarchy based on human ratings

The concept of animacy goes beyond the simple biological distinction of "alive" versus "dead." Human perception of animacy is influenced by several factors, including human-likeness and the ability to move, think, or reproduce (VanArsdall et al., 2017; VanArsdall and Blunt, 2022). These cognitive factors contribute to a scalar hierarchy of animacy, where living beings, particularly animals and humans, are perceived as more animate than others.

Radanović et al. (2016) conducted semantic rating experiments with 126 English-speaking college students, asking them to rate 72 nouns based on their perceived level of animacy. The results showed that both animals and human-denoting nouns received high ratings, with average scores exceeding 95 on a 100-point scale for animals like *dog*, *giraffe*, *cow*, and *squirrel* as well as for human-denoting nouns like *baby*, *mother*, and *girlfriend*. Interestingly, some human-denoting nouns like *teacher*, *prince*, and *queen* received slightly lower ratings (around 90), comparable to creatures like *worm*, *spider*, and *fly*. This indicates that, from a cognitive perspective, humans and animals are similarly animate, with certain animals even out-ranking humans.

A more recent large-scale online experiment by VanArsdall and Blunt (2022), with 1,500 native speakers of English rating 1,200 concrete nouns, found similar trends. Animals, especially mammals and birds, were rated higher than humans in terms of being alive, having the ability to reproduce, and their likelihood of movement. These findings support a human perception-based animacy hierarchy, where animals are perceived as at least as animate as humans, if not more so (see Eq. 1).

$$\text{Animals} > \text{Humans} > \text{Inanimate Entities} \quad (1)$$

### 2.3 Animacy hierarchy based on grammatical patterns

Animacy plays a crucial role in the grammatical systems of many languages. However, the way animacy is represented in grammar can vary. In some languages, animacy distinctions are discrete, with specific morphosyntactic markers assigned to nouns of different animacy levels, affecting person, number, case, and agreement (Corbett, 2006, 2012; Comrie, 1989; Croft, 1990; Ortmann, 1998; Santazilia, 2019, 2020; Silverstein, 1976; de Swart

et al., 2008, a.o.) For example, Table 1 illustrates how plural marking in Wambaya varies depending on animacy (Aguas, 1968:5-6; cited in Santazilia, 2020:p.823). Other languages exhibit gradient effects, where animacy influences the likelihood of one construction being chosen over another. For example, in English, when the indirect object (recipient) of a ditransitive verb is animate, the double-object construction (V NP NP) is more likely than the prepositional dative construction (V NP PP) (Bresnan et al., 2007). Similarly, with animate nouns, the *s*-genitive (*John's book*) is preferred over the *of*-genitive (*the book of John*) (Rosenbach, 2008; Stefanowitsch, 2003). When the object of a verb is more animate than the subject, passive voice is favored over active voice (Harris, 1978).

Although languages typically categorize nouns into two or three levels of animacy (e.g. humans vs. non-humans, living creatures vs. non-living entities), the exact cut-off points for these categories differ (Comrie, 1989; Santazilia, 2020; de Swart et al., 2008, a.o.) Cross-linguistic studies have synthesized these patterns into a generalized animacy hierarchy (e.g. Gardelle and Sorlin, 2018). One widely recognized version of this hierarchy is presented in Corbett (2000), where entities that bear more similarities with the self (i.e. the speaker) are considered more animate than others (the egophoricity principle, Dahl, 2008; Langacker, 1991; Yamamoto, 1999): see Eq. 2

$$\begin{aligned} &\text{Speaker (1}^{\text{st}} \text{ person pronoun)} > \\ &\text{Addressee (2}^{\text{nd}} \text{ person pronoun)} > \text{3}^{\text{rd}} \text{ person} > \\ &\text{Kin} > \text{Human} > \text{Animate} > \text{Inanimate} \end{aligned} \quad (2)$$

It is worth noting that grammatical animacy is assumed to be a universal feature (Dahl and Fraurud, 1996; Jespersen, 1924; Whaley, 1997, a.o.), even if its effects are gradient. A case in point is the optional marking of the direct object in Turkish. When sentences contain a ditransitive verb and an optional marker on the direct object, they become progressively less felicitous as the animacy of the direct object decreases. Sentences with the optional marker on human-denoting nouns receive higher acceptability ratings than those with the marker on animals or inanimate objects, even though the nouns denoting humans, animals and inanimate entities are all direct objects in those sentences (Krause and von Heusinger, 2019).

The universality of grammatical animacy suggests that the animacy hierarchy in Eq. 2, based on

cross-linguistic grammatical patterns, should also apply to English. This sets the stage for comparisons with the human perception-based hierarchy derived from English data, as introduced in the previous section. The two hierarchies exhibit different rankings, which we elaborate on in the next section.

## 2.4 Contrasting hierarchies

There are two key points of divergence between the hierarchy based on grammatical patterns and the one based on human perception. First, the grammar-based hierarchy ranks humans above other animate entities, whereas the human rating-based hierarchy often ranks animals as equally or even more animate than humans. Second, the grammar-based hierarchy makes finer distinctions among human-denoting nouns. For instance, in the grammar-based hierarchy, pronouns are ranked higher than human-denoting common nouns, and within pronouns, first-person pronouns are ranked higher than second-person pronouns, which in turn rank higher than third-person pronouns. In contrast, the rating-based hierarchy does not have principled predictions regarding the relative rankings among various human-denoting nouns.

These discrepancies raise important questions about the trilateral relationship between language use, cognition, and grammar. To what extent does the assumption that language use—specifically co-occurrence patterns—reflects human perceptions (Harris, 1954; Firth, 1957; Wittgenstein, 1953, a.o.) hold for ‘animacy’? What are the origins of grammatical animacy? Could grammatical patterns of animacy deviate from general trends in language use? Understanding how distributionally derived animacy hierarchies align with or diverge from perception-based and grammar-based hierarchies can provide valuable insights into the cognitive and linguistic foundations of animacy distinctions.

## 3 This study

In this study, we apply the semantic projection method to investigate the animacy ranking as reflected in English word embeddings. Specifically, we examine the divergent predictions made by the grammar-based hierarchy and the human rating-based hierarchy, focusing on three relative rankings: the rankings 1) between animals and humans; 2) between human-denoting common nouns and pronouns; and 3) among pronouns.

### 3.1 Experiment setup

**Selection of target words** We employed two sets of target words. The first set comprises three lists of 50 common nouns, categorized as humans (e.g. *artist, man, mother*), animals (e.g. *bird, cat, fish*), and objects (e.g. *cave, hill, rock*), respectively. These nouns were selected from the top 60 high-frequency words in each category from WordNet (Miller, 1995), following three selection criteria: (1) preference for high-frequency words over low-frequency ones; (2) inclusion of words that represent typical members of their respective categories to maintain clear contrasts among humans, animals, and objects; and (3) inclusion of both singular and plural forms to mitigate potential number-related effects. For example, although humans are technically animals, we excluded *human being* from the list of animals to preserve the distinction between the "humans" and "animals" categories. Another example related to the second criterion is *god*. While it could be argued to denote a human-like being, it is equally reasonable to classify *god* as an imaginary (and thus abstract) entity, making it more akin to an object. Such words were excluded from our analyses.

The second set of target words includes 31 words in total, consisting of singular and plural first-, second-, and third-person pronouns. This set includes nominative (e.g., *I*), accusative (e.g., *me*), adjectival possessive (e.g., *my*), possessive (e.g., *mine*), and reflexive (e.g., *myself*) pronouns. Inclusion of pronouns beyond the nominative forms provides a more balanced comparison between pronouns and common nouns.

**Selection of contexts** Vulić et al. (2020) demonstrated that averaging over 10 contexts is sufficient for capturing type-level lexical information, though increasing the number of contexts improves performance only marginally. In line with this, we sampled 30 sentences for each target word from the Brown Corpus (Francis and Kucera, 1979), accessed via the Natural Language Toolkit (NLTK) (Bird et al., 2009) in Python. In cases where fewer than 30 sentences were available, all sentences were used. Some nouns in the common noun set were excluded from subsequent analyses due to the unavailability of sufficient contexts (e.g., *antelopes, goats, grasshopper*). Additionally, we filtered out cases where nouns were used as denominalized verbs (e.g., *to duck a punch*) by restricting the part-of-speech (POS) tags of target words in the first set

to NN or NNS (singular or plural nouns).<sup>1</sup>

For the pronoun set, we manually verified the sampled sentences to exclude non-pronoun uses. For instance, sentences where *mine* refers to a bomb or mineral extraction (e.g., *coal mine*) were excluded. Additionally, since pronouns like "you" have multiple meanings (i.e., second person singular or plural), we did not attempt to differentiate between such polysemous uses, as those distinctions are orthogonal to the animacy feature we aim to study.

**Embedding models** Previous studies have shown that embeddings differ qualitatively across models and layers (Ethayarajh, 2019). To understand whether specific patterns of animacy ranking are model-dependent or restricted to particular layers, we employed two pre-trained embeddings models: BERT and GPT-2. The models are chosen for their contrasting architectures:

- BERT (Bidirectional Encoder Representations from Transformers) is a bidirectional transformer model designed for language understanding, which captures relationships from both left and right contexts (Devlin et al., 2019).<sup>2</sup>
- GPT-2 is a unidirectional transformer decoder, designed for language generation tasks (Radford et al., 2019).<sup>3</sup> It processes language sequentially from left to right.

We extracted embeddings from all twelve internal layers of both models using PyTorch and the Hugging Face transformers library. This allows us to determine how patterns of animacy evolve across layers in these two models.

**Extraction of embeddings and derivation of animacy scores** For each target word, we extracted

<sup>1</sup>Although our corpus may seem smaller than those typically used in computational linguistics studies, it is comparable in size to corpora from some previous studies. Radanović et al. (2016) analyzed 72 English nouns of various categories, only about 13 of which were human-denoting nouns (*baby, mother, girlfriend, boy, uncle, sailor, brother, professor, teacher, prince, queen, witch, cook*). In contrast, our study, after filtering, analyzes 170 words in total, including 46 human-denoting nouns. Other categories (animals, pronouns, objects) are of roughly comparable sizes.

<sup>2</sup><https://huggingface.co/google-bert/bert-base-uncased>, accessed and used under Apache License Version 2.

<sup>3</sup><https://huggingface.co/openai-community/gpt2>, accessed and used under MIT License.



its embeddings in context.<sup>4</sup> If a target word was out-of-vocabulary,<sup>5</sup> we used the mean of its sub-word token embeddings. The embeddings for each target word were averaged across contexts at each layer.

To derive the vectors representing the semantic feature of animacy, we averaged, at each layer, the embeddings of animal and object categories separately (Eq. 3, 4), then subtracted the average embeddings of objects from animals (Eq. 5).

$$\mathbf{v}_{animal}^l = \frac{1}{N_{animal}} \sum_{i=1}^{N_{animal}} \mathbf{e}_{animal,i}^l \quad (3)$$

$$\mathbf{v}_{object}^l = \frac{1}{N_{object}} \sum_{i=1}^{N_{object}} \mathbf{e}_{object,i}^l \quad (4)$$

$$\mathbf{a}^l = \mathbf{v}_{animal}^l - \mathbf{v}_{object}^l \quad (5)$$

Here,  $\mathbf{e}_{animal,i}^l$  and  $\mathbf{e}_{object,i}^l$  denote the embeddings of the  $i$ -th animal or object word on Layer  $l$ .  $N_{animal}$  and  $N_{object}$  are the number of words in the animal and object categories.

These vectors  $\mathbf{a}^l$  were transformed into unit vectors (Eq. 6). The animacy score  $s_w^l$  for a target word  $w$  on Layer  $l$  was calculated by projecting its embeddings  $\mathbf{e}_w^l$  onto the (extensions of the) layer-specific unit vector  $\hat{\mathbf{a}}^l$  (Eq. 7).

$$\hat{\mathbf{a}}^l = \frac{\mathbf{a}^l}{\|\mathbf{a}^l\|} \quad (6)$$

$$s_w^l = \mathbf{e}_w^l \cdot \hat{\mathbf{a}}^l \quad (7)$$

**Statistical analyses** Given that the animacy scores did not follow a normal distribution, we applied non-parametric tests to analyze the data. Specifically, we used the Kruskal-Wallis tests and performed post-hoc Dunn’s tests with Benjamini-Hochberg correction (Benjamini and Hochberg, 1995) to account for multiple comparisons. These statistical analyses were conducted using R (R Core Team, 2024) with the `dunn.test` package (Dinno, 2024). The results are reported in the next sections.

## 3.2 Experiment 1 & 2

Fig. 1 and 2 show the embedding-derived animacy scores for all target words used in analyses. Words of different categories exhibit distinct patterns despite within-category variation. The distinctions appear to be more prominent in BERT than in GPT-2.

We conducted a Kruskal-Wallis test to compare the animacy scores among nouns denoting animals, humans, objects, and pronouns across all layers in both embedding models. Significant differences were found between categories across all layers in both models ( $p < 0.05$  for all comparisons). Below, we present the results of post-hoc Dunn’s tests. Experiment 1 focuses on the relative rankings among humans, animals, and objects, while Experiment 2 compares human-denoting common nouns and pronouns.

### 3.2.1 Experiment 1: humans, animals, objects

By-category mean and standard deviation of words denoting animals, humans and objects are shown in Fig. 3 (and Table 3 in Appendix A.2). Post-hoc Dunn’s tests revealed that in BERT, animals consistently ranked higher than humans and objects across all layers of BERT (animals > humans > objects,  $p < 0.05$  for all). In GPT-2, all layers except Layer 12 followed this pattern, with the differences between animals and humans achieving significance on Layers 1-2 and 10-11, but not on intermediate Layers 3-9 ( $p < 0.05$ ), and the differences between humans and objects achieving significance on Layers 1-11 ( $p < 0.05$ ). Interestingly, Layer 12 in GPT-2 exhibited a reverse pattern where objects were ranked higher than humans, but the differences were not statistically significant.

The finding that animals rank higher than humans contradicts predictions from the grammar-based animacy hierarchy but aligns with the human-rating-based hierarchy. This confirms that the semantic projection approach, as introduced in Grand et al. (2022), is effective in capturing perceived animacy in a human-like manner.

<sup>4</sup>The code we used can be found at <https://github.com/yesvivan/animacy>

<sup>5</sup>Which words were out-of-vocabulary would depend on the tokenization method used in the models. BERT uses Word Piece vocabulary; words like *anaconda* and *bumblebee* were out-of-vocabulary and were represented as the combination of multiple sub-word pieces. GPT2 uses BPE-based tokenization; words like *snake* and *sheep*, which were in-vocabulary in BERT, were multi-token (i.e. out-of-vocabulary) words in GPT2.

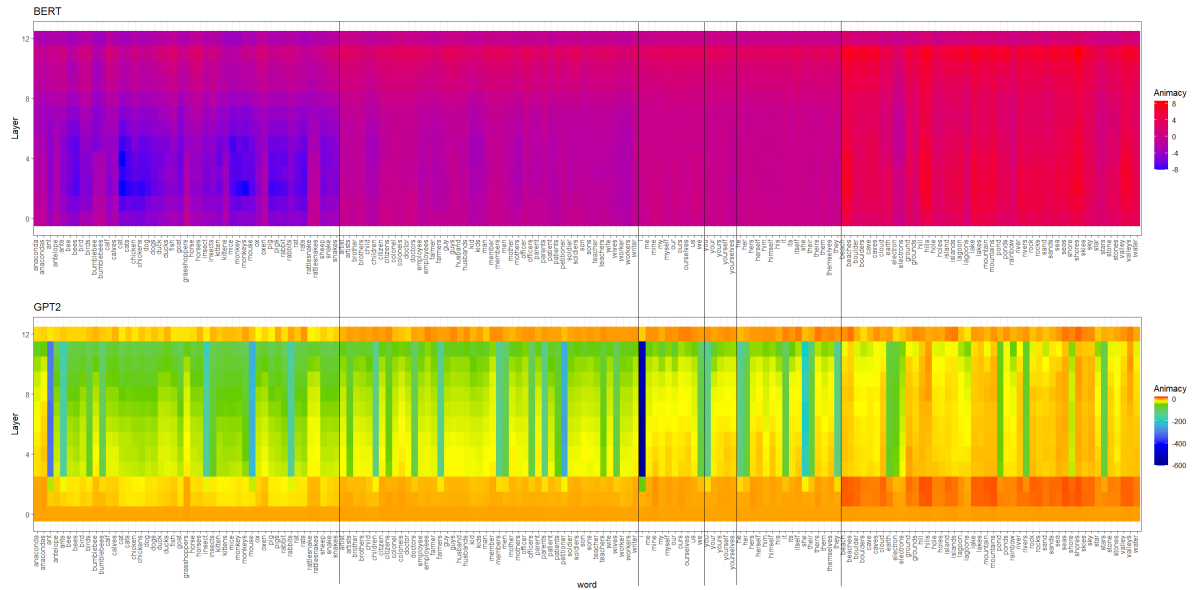


Figure 1: Heatmap showing animacy scores for all words used in the analyses across all layers in BERT (top) and GPT-2 (bottom). The vertical black lines in both panels demarcate different word categories, arranged from left to right: animals, humans, first-person pronouns, second-person pronouns, third-person pronouns, and objects. In both panels, the blue end of the color spectrum represents the highest level of animacy. The full list of words analyzed can be found in Appendix A.1.

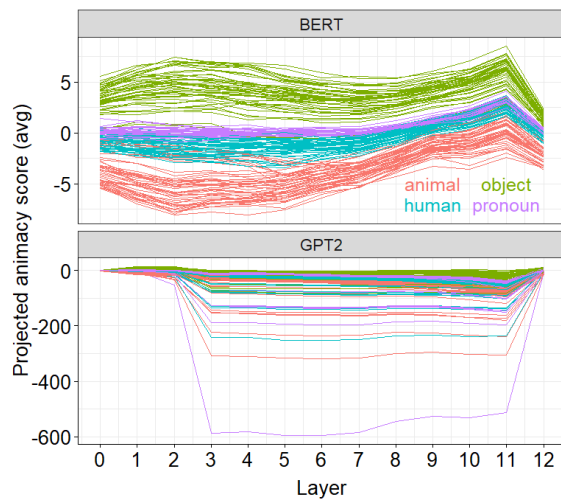


Figure 2: Line graph showing animacy scores of all target words used in the analyses across all layers in BERT (top) and GPT-2 (bottom). Categories (animals, humans, pronouns, objects) are color-coded. Within each panel, the lower the score, the higher the animacy.

### 3.2.2 Experiment 2: pronouns vs. human-denoting common nouns

Next, we examined the animacy scores for human-denoting common nouns and pronouns. Results from Dunn's tests indicated that human-denoting common nouns were generally ranked higher than pronouns. The differences were statistically significant ( $p < 0.05$ ) across most layers in BERT,

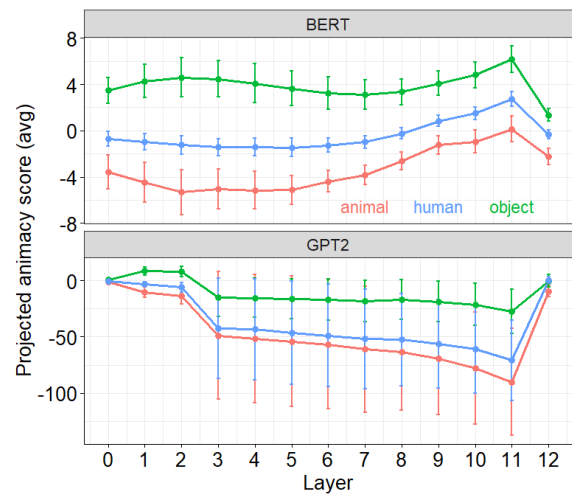


Figure 3: Line graph showing mean and standard deviation (error bars) of animacy scores of categories animals, humans and objects, averaged across words. Within each panel, the lower the score, the higher the animacy.

except Layers 9-11. In GPT-2, the ranking was the same, with common nouns ranking higher than pronouns, though the differences were significant only in Layer 1 and Layers 3-5 ( $p < 0.05$ ).

Again, these findings challenge the grammar-based hierarchy, which predicts that pronouns should rank higher than human-denoting common nouns.

### 3.3 Experiment 3: pronouns

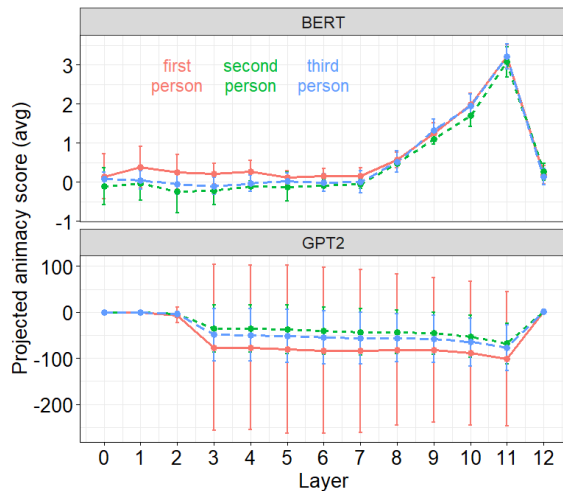


Figure 4: Line graph showing mean and standard deviation (error bars) of animacy scores of first-, second-, and third- person pronouns, averaged across words. Within each panel, the lower the score, the higher the animacy.

We analyzed the personal pronoun series (mean and standard deviations shown in Fig. 4, and in Table 4 in Appendix A.2) to test the predictions from the grammar-based animacy hierarchy under the egophoricity principle. Kruskal-Wallis tests showed no significant differences between first-, second-, and third- person pronouns across most layers of BERT and GPT-2,<sup>6</sup> except for Layers 3, 4, and 6 in BERT. Post-hoc Dunn’s tests revealed that on Layers 3 and 4 in BERT, second-person pronouns ranked higher than third-person pronouns and first-person pronouns. The differences between second- and third- person pronouns were not statistically significant, but both were significantly higher than first-person pronouns ( $p < 0.05$ ). On Layer 6, the same ranking emerged; however, pairwise comparisons did not achieve statistical significance.

In summary, while grammatical hierarchies based on the egophoricity principle rank first-person pronouns as the most animate, our embedding-based analysis revealed a different ranking. On layers that showed differentiation, second- and third person pronouns were ranked as more animate than first-person pronouns.

<sup>6</sup>In GPT-2 Layers 3-11, although the mean animacy scores of first-person pronouns were lower than those of second- and third-person pronouns, the differences were not statistically significant due to the high variability among first-person pronouns.

## 4 General Discussions

### 4.1 Embedding-derived animacy

**Semantic projection approach is effective** The analyses reveal that animacy rankings derived from word embeddings align well with the human-ranking-based hierarchy. Specifically, our results show that common nouns denoting animals are ranked higher in animacy than humans, which replicates findings from behavioral surveys. These findings provide further evidence that embedding-derived metrics of lexical semantics can serve as reliable proxies for human perception of semantic features, such as animacy. This adds to the growing support for distributional approaches to lexical semantics (Harris, 1954; Firth, 1957; Mikolov et al., 2013; Miller and Charles, 1991; Pennington et al., 2014). More broadly, these findings contribute to the case for using language models as effective proxies for human-like agents in various linguistic tasks (e.g. Andreas, 2022).

**Mismatches between embedding-derived animacy and grammatical animacy** Our results deviate from predictions from grammar-based animacy hierarchy. Among common nouns denoting animals, objects, and humans, embedding-derived animacy scores ranked humans as less animate than animals; pronouns were ranked as less animate than human-denoting common nouns and first-person pronouns were overall not more animate than second- or third- person pronouns. These findings contradict the egophoricity principle (Dahl, 2008).<sup>7</sup>

Our findings bear implications for understanding the sources of grammatical animacy. The fact that grammar-based hierarchy imposes relative rankings (e.g. between common nouns for animals and humans) that are different from those in human perception and general language use (as reflected in averaged embeddings across contexts) suggests that language learners, when acquiring the grammar, must defy the general tendencies in perception and language use, and selectively focus on a subset of constructions (e.g. ditransitives, Krause and von Heusinger, 2019), to develop alternative ranking preferences. Thus, our findings provide evidence

<sup>7</sup>Note, however, although as a category, first-person pronouns were not more animate than second-, and third-person pronouns, the first-person singular pronoun in nominative form *I* was perhaps the most animate word across Layers 3-11 in GPT-2, which is consistent with the egophoricity principle (Dahl, 2008).

that ‘animacy’, as a grammatical feature, is shaped by inductive biases, highlighting the intricate interplay between cognitive perception, linguistic usage, and grammatical representation.

## 4.2 Capturing semantic features in neural network models

**Bidirectional vs. unidirectional context** In our study, we utilized both a bidirectional model, BERT, and a unidirectional model, GPT-2. Remarkably, both models produced comparable results, yielding identical rankings across almost all internal layers for common nouns in the categories of animals, humans, and objects. This suggests that the left-context used by GPT-2 and the bidirectional contexts employed by BERT convey similar information when encoding the semantic feature of animacy. Hence, using unidirectional context alone is sufficient for inferring relative animacy ranking that align with human perception. This contrasts the task of inferring fine-grained animacy distinctions with tasks like constituency parsing, which rely more heavily on bi-directional context (e.g. Li and Risteski, 2021). It may be that the inference of semantic features generally relies less on right-side context compared to the inference of syntactic features. Future studies could further explore this topic by systematically examining the utility of uni- vs. bi-directional contexts across a diverse range of tasks.

### Information encoding across internal layers

Our results align with previous findings suggesting that lower layers of embedding models predominantly encode type-level information, while higher layers capture context-specific characteristics (Ethayarajh, 2019; Tenney et al., 2019). In our findings, differences in animacy diminished at higher levels, particularly at Layer 12. Since abstract type-level information is less likely to change across contexts, and that higher layers tend to enhance the representation of contextually dependent features, the observed reduction in animacy differences between common nouns for humans, animals, and objects, as well as among first-, second-, and third-person pronouns, is not unexpected.

**Within-category variations** Since the focus of our study is on the comparisons of relative animacy rankings across categories of words, we did not conduct in-depth analyses on within-category variations or same-word cross-layer differences in

animacy scores. However, some notable patterns are worth mentioning.

First, substantial within-category variation is present in the animacy scores derived from both BERT and GPT-2. For example, in both models, *ant* was more animate than other animal words, whereas *rattlesnake(s)* and *grasshopper* were comparatively less animate (see Fig. 1).

Second, within-category variation appears more pronounced in GPT-2. A particularly striking case is the first-person singular nominative pronoun *I*, which, in GPT-2 Layers 3-11, emerged as the most animate word across all words. In contrast, BERT did not exhibit a similar distinction for *I* relative to other pronouns. Beyond capturing the animacy of the entities denoted by words, GPT-2’s embedding-derived animacy scores also seemed systematically sensitive to additional grammatical and semantic features. For instance, nominative case pronouns consistently appeared more animate than other pronouns. Moreover, plural-singular noun pairs exhibited marked differences: *pond* was more animate than *ponds*, and *stars* was more animate than *star*. Such differences were much less drastic in BERT.

Additionally, we observed differences between the two models in how animacy for the same word was encoded across layers. In GPT-2, animacy scores for highly animate words (e.g., *I* and *ant*) remained relatively stable across Layers 3-11, with noticeable shifts occurring primarily from Layer 2 to Layer 3 and from Layer 11 to Layer 12. In contrast, BERT exhibited a more dynamic encoding of animacy, with incremental changes in animacy scores from Layer 0 to Layer 11, except for the least animate words (e.g., *beach* and *beaches*), whose scores remained relatively stable.

These observations highlight the nuanced ways in which animacy was encoded across words (even within the same broad category) and across layers and models. They suggest two insights: 1) semantic projection may offer a viable approach for exploring sub-dimensions of animacy; 2) while both BERT and GPT-2 consistently ranked major categories (e.g., animals, humans, pronouns, objects) in similar relative orders, GPT-2 appears to be more sensitive to grammatical and semantic features.

## 5 Conclusion

In this study, we applied the semantic projection approach to animacy, a feature that has not been previously explored using this method. We com-



pared the relative animacy rankings of nouns denoting animals, humans, objects, and first-, second-, and third-person pronouns, as derived from word embeddings, with rankings derived from human behavioral ratings of animacy and from grammatical patterns. Our results support the semantic projection approach as an effective method for deriving proxies of human perception from word embeddings and offer insights into the sources of grammatical animacy.

## 6 Limitations and future directions

While our study provides valuable insights, several limitations warrant acknowledgement. First, our experiments focused exclusively on English nouns. Future research could extend embedding-based animacy analyses to other languages, particularly those with morphosyntactic markings of animacy, to assess the cross-linguistic validity of the patterns observed here.

Second, our analysis of embedding-derived animacy scores did not account for factors such as sentence length, the semantic roles of target words, or their syntactic dependency relations. We assumed that averaging across 10 contexts would sufficiently capture type-level lexical information (Vulić et al., 2020). Future research could examine whether these factors systematically influence derived animacy scores.

Although our study does not have immediate practical applications, our finding that embedding-derived animacy closely aligns with human rating-based animacy rankings suggests that embeddings may be useful for automatic animacy classification (e.g., Tepei and Bloem 2024). However, tasks requiring animacy classification vary in nature: some target semantic or cognitive animacy (e.g., atypical animacy in metaphors; Coll Ardanuy et al., 2020; Hanna et al., 2023; Jahan et al., 2018), while others focus on grammatical animacy (e.g., translating from languages without morphosyntactic animacy marking to those with such marking). A promising direction for future research is to identify representations of animacy in language models' latent space that align with grammar-based animacy hierarchies or to develop methods (e.g., fine-tuning on grammatical tasks) to induce 'biases'—akin to human inductive biases—in language models, thereby shifting their embeddings toward greater alignment with grammar-based animacy hierarchies.

Lastly, while our findings indicate that the se-

mantic projection approach effectively approximates human perceptions of overall animacy, future studies could investigate specific sub-dimensions of animacy, such as mobility, where relative rankings may diverge from overall animacy rankings. Further exploration of these subcomponents using the semantic projection method could enhance our understanding and modeling of complex, nested semantic structures. Additionally, examining how embedding-derived animacy scores correlate with human ratings and whether distinct layers correspond to different sub-dimensions (à la Hollis and Westbury, 2016) would be a valuable avenue for future research.

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## A Appendix

### A.1 Experiment materials: words

| Animals    |              |            |              |
|------------|--------------|------------|--------------|
| anaconda   | calves       | horse      | pig          |
| anacondas  | cat          | horses     | pigs         |
| ant        | cats         | insect     | rabbit       |
| antelope   | chicken      | insects    | rabbits      |
| ants       | chickens     | kitten     | rat          |
| bee        | dog          | kittens    | rats         |
| bees       | dogs         | mice       | rattlesnake  |
| bird       | duck         | monkey     | rattlesnakes |
| birds      | ducks        | monkeys    | sheep        |
| bumblebee  | fish         | mouse      | snake        |
| bumblebees | goat         | ox         | snakes       |
| calf       | grasshoppers | oxen       |              |
| Humans     |              |            |              |
| artist     | employee     | members    | soldiers     |
| artists    | employees    | men        | son          |
| brother    | farmer       | mother     | sons         |
| brothers   | farmers      | mothers    | teacher      |
| child      | guy          | officer    | teachers     |
| children   | guys         | officers   | wife         |
| citizen    | husband      | parent     | wives        |
| citizens   | husbands     | parents    | worker       |
| colonel    | kid          | patient    | workers      |
| colonels   | kids         | patients   | writer       |
| doctor     | man          | petitioner |              |
| doctors    | member       | soldier    |              |
| Objects    |              |            |              |
| beach      | hill         | pond       | shores       |
| beaches    | hills        | ponds      | skies        |
| boulder    | hole         | rainbow    | sky          |
| boulders   | holes        | river      | star         |
| cave       | island       | rivers     | stars        |
| caves      | islands      | rock       | stone        |
| cloud      | lagoon       | rocks      | stones       |
| earth      | lagoons      | sand       | valley       |
| electron   | lake         | sands      | valleys      |
| electrons  | lakes        | sea        | water        |
| ground     | mountain     | seas       |              |
| grounds    | mountains    | shore      |              |
| Pronouns   |              |            |              |
| I          | us           | her        | itself       |
| me         | we           | hers       | she          |
| mine       | you          | herself    | their        |
| my         | your         | him        | theirs       |
| myself     | yours        | himself    | them         |
| our        | yourself     | his        | themselves   |
| ours       | yourselves   | it         | they         |
| ourselves  | he           | its        |              |

Table 2: Words used in experiments.



## A.2 Descriptive statistics

| Layer | BERT    |       |        |      |          |      |         |      | GPT-2   |       |        |       |          |        |         |       |
|-------|---------|-------|--------|------|----------|------|---------|------|---------|-------|--------|-------|----------|--------|---------|-------|
|       | Animals |       | Humans |      | Pronouns |      | Objects |      | Animals |       | Humans |       | Pronouns |        | Objects |       |
|       | Mean    | SD    | Mean   | SD   | Mean     | SD   | Mean    | SD   | Mean    | SD    | Mean   | SD    | Mean     | SD     | Mean    | SD    |
| 0     | -3.57   | -0.71 | 1.48   | 0.62 | 0.07     | 0.39 | 3.47    | 1.14 | -0.9    | 0.33  | -0.33  | 0.14  | -0.17    | 0.14   | 0.62    | 0.32  |
| 1     | -4.48   | -0.96 | 1.73   | 0.71 | 0.14     | 0.41 | 4.27    | 1.43 | -10.41  | 4.45  | -2.84  | 1.48  | -0.53    | 1.24   | 8.35    | 3.73  |
| 2     | -5.32   | -1.23 | 1.96   | 0.79 | 0.01     | 0.4  | 4.61    | 1.67 | -13.85  | 6.99  | -6.04  | 4.31  | -4.47    | 9.99   | 8.16    | 4.57  |
| 3     | -5.04   | -1.44 | 1.72   | 0.74 | -0.02    | 0.3  | 4.47    | 1.56 | -48.8   | 56.71 | -42.5  | 44.83 | -55.64   | 109.55 | -14.96  | 16.95 |
| 4     | -5.15   | -1.42 | 1.61   | 0.77 | 0.05     | 0.27 | 4.09    | 1.7  | -51.69  | 56.76 | -43.36 | 44.83 | -55.91   | 108.41 | -15.33  | 17.18 |
| 5     | -5.13   | -1.45 | 1.25   | 0.81 | 0.03     | 0.23 | 3.64    | 1.51 | -53.96  | 57.82 | -46.28 | 45.88 | -58.53   | 110.76 | -16.36  | 17.8  |
| 6     | -4.38   | -1.25 | 0.92   | 0.61 | 0.03     | 0.22 | 3.24    | 1.41 | -56.61  | 57.47 | -48.7  | 45.48 | -61.59   | 109.85 | -16.94  | 18.23 |
| 7     | -3.82   | -0.99 | 0.84   | 0.52 | 0.04     | 0.24 | 3.1     | 1.29 | -60.95  | 56.11 | -51.91 | 44.38 | -63.52   | 107.54 | -18.26  | 18.34 |
| 8     | -2.63   | -0.23 | 0.75   | 0.49 | 0.53     | 0.22 | 3.35    | 1.1  | -63.33  | 51.97 | -52.57 | 41.06 | -61.88   | 99.83  | -16.85  | 17.87 |
| 9     | -1.25   | 0.84  | 0.78   | 0.49 | 1.26     | 0.26 | 4.1     | 1.08 | -69.13  | 50    | -56.46 | 39.34 | -63.71   | 95.69  | -18.62  | 17.87 |
| 10    | -0.95   | 1.49  | 1      | 0.57 | 1.92     | 0.29 | 4.8     | 1.08 | -77.86  | 49.99 | -61.11 | 38.91 | -70.69   | 95.8   | -21.25  | 18.77 |
| 11    | 0.15    | 2.74  | 1.12   | 0.61 | 3.19     | 0.32 | 6.19    | 1.16 | -89.98  | 47.77 | -70.88 | 35.94 | -84.1    | 89.71  | -27.23  | 19.5  |
| 12    | -2.24   | -0.34 | 0.73   | 0.38 | 0.17     | 0.21 | 1.36    | 0.55 | -9.47   | 4.91  | -0.15  | 3.29  | 1.68     | 3.65   | -0.43   | 5.46  |

Table 3: Mean and standard deviations of target words by category: animals, humans, pronouns, objects, in BERT (left) and GPT-2 (right).

| Layer | BERT  |      |        |      |       |      | GPT-2   |        |        |       |        |       |
|-------|-------|------|--------|------|-------|------|---------|--------|--------|-------|--------|-------|
|       | First |      | Second |      | Third |      | First   |        | Second |       | Third  |       |
|       | Mean  | SD   | Mean   | SD   | Mean  | SD   | Mean    | SD     | Mean   | SD    | Mean   | SD    |
| 0     | 0.14  | 0.57 | -0.11  | 0.48 | 0.08  | 0.18 | -0.15   | 0.15   | -0.25  | 0.12  | -0.16  | 0.13  |
| 1     | 0.38  | 0.53 | -0.04  | 0.42 | 0.04  | 0.24 | -0.56   | 1.72   | -0.6   | 0.81  | -0.48  | 1.07  |
| 2     | 0.25  | 0.44 | -0.25  | 0.54 | -0.06 | 0.25 | -6.15   | 16.36  | -2.95  | 4.87  | -3.89  | 5.44  |
| 3     | 0.22  | 0.27 | -0.23  | 0.36 | -0.11 | 0.22 | -76.54  | 180.76 | -35.88 | 51.73 | -48.75 | 56.67 |
| 4     | 0.26  | 0.28 | -0.11  | 0.14 | -0.04 | 0.21 | -76.83  | 178.62 | -35.2  | 51.5  | -49.31 | 56.35 |
| 5     | 0.11  | 0.18 | -0.12  | 0.37 | 0.02  | 0.2  | -80.29  | 182.42 | -36.97 | 52.6  | -51.68 | 57.56 |
| 6     | 0.16  | 0.19 | -0.09  | 0.14 | -0.02 | 0.23 | -82.86  | 180.93 | -40.39 | 51.56 | -54.93 | 57.32 |
| 7     | 0.15  | 0.21 | -0.06  | 0.11 | 0     | 0.28 | -84.42  | 177.14 | -42.91 | 50.35 | -56.9  | 56.09 |
| 8     | 0.56  | 0.19 | 0.48   | 0.05 | 0.52  | 0.27 | -81.33  | 164.13 | -43.01 | 47.1  | -55.63 | 52.59 |
| 9     | 1.24  | 0.27 | 1.1    | 0.14 | 1.32  | 0.28 | -82.22  | 156.84 | -45.83 | 45.43 | -57.73 | 51.29 |
| 10    | 1.99  | 0.27 | 1.69   | 0.27 | 1.95  | 0.29 | -89.15  | 156.55 | -52.84 | 45.89 | -64.73 | 52.13 |
| 11    | 3.22  | 0.3  | 3.07   | 0.38 | 3.21  | 0.33 | -101.87 | 145.87 | -68.37 | 44.32 | -77.91 | 49.74 |
| 12    | 0.18  | 0.23 | 0.26   | 0.22 | 0.14  | 0.2  | 1.73    | 4.39   | 1.8    | 1.71  | 1.61   | 3.79  |

Table 4: Mean and standard deviations of target words in the pronoun set: first-, second-, and third person pronouns, in BERT (left) and GPT-2 (right).